

LAND APPLICATION - VECTOR ATTRACTION REDUCTION

FACILITY: Cadillac WWTF

CITY: Cadillac MI

How were the vector attraction reduction requirements met?

Site Identification:

Are the site identification for this form the same as previously identified on the site description form?

☒ Yes ☐ No If no, provide a correct site description form for each site and attach it to this form.

CHECK ALL THAT APPLY.

NOTE: ONLY ONE (1) OF THE VECTOR ATTRACTION REDUCTION REQUIREMENTS MUST BE MET.

- 1) ☐ The mass of volatile solids in the sewage sludge was reduced by a minimum of 38 percent.

$$\text{Volatile Solids Reduction} = \frac{\text{VS in} - \text{VS out}}{\text{VS in} - (\text{VS in} \times \text{VS out})} \times 100$$

$$\text{Volatile Solids Reduction} = \underline{\hspace{2cm}} \%$$

VS in and VS out in decimal, e.g. 50% = 0.50

- 2) ☐ When the 38 percent volatile solids reduction requirement could not be met for an anaerobically digested sewage sludge, vector attraction reduction was demonstrated by digesting a portion of the previously digested sewage sludge anaerobically in the laboratory in a bench-scale unit for 40 additional days at a temperature between 30 and 37 C (90 to 99 F). At the end of the 40 days, the volatile solids in the sewage sludge at the beginning of that period was reduced by less than 17 percent.

$$\text{Volatile Solids Reduction} = \underline{\hspace{2cm}} \%$$

- 3) ☐ When the 38 percent volatile solids reduction requirement could not be met for an aerobically digested sewage sludge, vector attraction reduction was demonstrated by digesting a portion of the previously digested sewage sludge that has a percent solids of two (2) percent or less aerobically in the laboratory in a bench-scale unit for 30 additional days at 20 C (68F). At the end of the 30 days, the volatile solids in the sewage sludge at the beginning of that period was reduced by less than 15 percent.

$$\text{Volatile Solids Reduction} = \underline{\hspace{2cm}} \%$$

- 4) ☐ The specific oxygen uptake rate (SOUR) for sewage sludge treated in an aerobic process was equal to or less than 1.5 milligrams of oxygen per hour per gram of total solids (dry weight basis) at a temperature of 20 C (68 F).

$$\text{Specific Oxygen Uptake Rate} = \underline{\hspace{2cm}} \text{ mg/hr/gram}$$

- 5) ☐ Sewage sludge was treated in an aerobic process for 14 days or longer. During that time, the temperature of the sewage sludge was higher than 40 C (104 F) and the average temperature of the sewage sludge was higher than 45 C (113 F).

- 6) _____ The pH of the sewage sludge was raised to 12 SU or higher by alkali addition and, without the addition of more alkali, remained at a pH of 12 SU or higher for two (2) hours and then at a pH of 11.5 SU or higher for an additional 22 hours.

Indicate the individual sites that receive alkali/lime treated sludge.

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

- 7) _____ The percent solids of sewage sludge that does not contain unstabilized solids generated in a primary wastewater treatment process was equal to or greater than 75 percent based on the moisture content and total solids prior to mixing with other materials.

- 8) _____ The percent solids of sewage sludge that contains unstabilized solids generated in a primary wastewater treatment process was equal to or greater than 90 percent based on the moisture content and total solids prior to mixing with other materials.

- 9) x Sewage sludge was injected below the surface of the land
No significant amount of the sewage sludge was present on the land surface within one (1) hour after the sewage sludge was injected.

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	<u> x </u>	<u> x </u>	<u> x </u>	<u> x </u>	<u> x </u>	<u> x </u>	<u> x </u>	<u> x </u>	_____	_____

- 10) _____ Sewage sludge applied to the land surface was incorporated into the soil within six (6) hours after application to the land.

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

CERTIFICATION

I certify under the penalty of law that the information as listed above is complete and accurate to the best of my knowledge. This determination has been made under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information used to determine the management practices have been met. I am aware that there are significant penalties for false certification, including the possibility of fine and imprisonment.

Todd Timmerman Supervisor
Name and Official Title (type or print)

Todd Timmerman
Signature

10-2-19
Date Signed